

B.Sc. (Part-II) Third Semester (NEP) Winter 2025
109700/110700/112700/123700/134700/135700 (Computer Application / Computer
Application (Voc) / Computer Science / Information Technology / Data Analytics /
Data Science & Analysis) - IKS in Computational Sciences

P. Pages : 1

Time : Two Hours



AI - 6025

Max. Marks : 30

- Notes :
1. All questions carry **six** marks each.
 2. Draw diagrams & write equations wherever necessary.

1. Write answers in one or two lines **any six**. 6
 - a) Define Indian Knowledge System.
 - b) What is Caturdasa-vidyasthana?
 - c) What do you mean by Sanskrit phonetics?
 - d) Define Natural Language Processing (NLP).
 - e) What is meant by Number System?
 - f) Define "Place value".
 - g) What is Binary Mathematics?
 - h) Who is Aryabhata?
2. Short answers **any two**. 6
 - a) Define Indian Knowledge System (IKS). Explain in brief.
 - b) What is Caturdasa-Vidyasthana? Explain brief.
 - c) State any two unique aspects of IKS.
 - d) What are components of a language? Explain in brief.
3. Long answer **any one**. 6
 - a) Explain the role of Sanskrit in Natural Language Processing.
 - b) Describe contributions of ancient Indian mathematicians in the field of algebra and geometry.
4. Short answers **any two**. 6
 - a) What is the place value system? Explain in detail.
 - b) Define the concept of zero and its importance. Explain in detail.
 - c) What are large numbers? Give examples from Indian numeral history.
 - d) Explain salient features of the Indian numeral system.
5. Long answer **any one**. 6
 - a) Explain ancient Indian methods of measuring time, distance, and weight.
 - b) Describe the concept of binary mathematics and its roots in ancient India.
